

Theoretical Analysis of Gravitational-Wave Signals in LIGO Data

A Theoretical and Data-Centric Framework for Gravitational-Wave Detection

Abstract

Gravitational waves provide a direct observational probe of the dynamical structure of spacetime and the most energetic astrophysical events in the Universe. The Laser Interferometer Gravitational-Wave Observatory (LIGO) detects these extremely weak perturbations through changes in the relative lengths of its interferometer arms.

This work develops a theoretical framework for studying gravitational wave signals in LIGO data, beginning with General Relativity and the propagation of spacetime perturbations and extending toward signal representation, detector noise, and computational classification.

Particular attention is given to the theoretical relations investigated in this study and to their possible connection with observable properties of gravitational-wave signals. The computational component further examines the ability of machine-learning models to distinguish rare astrophysical signals from instrumental noise.

The proposed theoretical relations are presented as hypotheses rather than established physical laws and are intended to provide a basis for further mathematical derivation, numerical simulation, and observational testing.

Keywords: gravitational waves, LIGO, General Relativity, spacetime, gravitational-wave detection, signal analysis, machine learning, instrumental noise.

1 Introduction

Gravitational waves are propagating perturbations of spacetime predicted by Einstein's General Theory of Relativity [1]. Their direct observation provides a fundamentally different method of studying the Universe because gravitational-wave observations probe the dynamics of massive compact objects rather than relying exclusively on electromagnetic radiation [4].

The Laser Interferometer Gravitational-Wave Observatory (LIGO) was designed to detect extremely small variations in spacetime strain [2]. These variations can originate from astrophysical systems such as binary black-hole mergers and neutron-star systems.

The measured detector data, however, do not consist exclusively of astrophysical signals. Instrumental and environmental disturbances produce a substantial background of noise and transient glitches. Consequently, gravitational-wave detection is simultaneously a physical and a signal-analysis problem [3].

The present study therefore considers three connected levels:

1. The theoretical description of spacetime and gravitational waves;
2. The physical and mathematical representation of the detected signal;
3. The computational distinction between astrophysical signals and instrumental noise.

The purpose is not to introduce a final or validated physical theory, but rather to construct a theoretical framework that can be subjected to mathematical and observational testing.

2 Theoretical Foundation

2.1 General Relativity

The theoretical description begins with Einstein's field equations,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}. \quad (1)$$

Here, $G_{\mu\nu}$ describes the geometry of spacetime, $g_{\mu\nu}$ is the metric tensor, Λ is the cosmological constant, and $T_{\mu\nu}$ represents the energy-momentum tensor.

For weak gravitational fields, the metric can be expressed as a small perturbation of the Minkowski metric,

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad (2)$$

where $h_{\mu\nu}$ represents the gravitational perturbation.

2.2 Gravitational-Wave Propagation

Under the weak-field approximation and an appropriate gauge condition, the gravitational perturbation satisfies a wave equation,

$$\square h_{\mu\nu} = 0. \quad (3)$$

A simplified plane-wave representation can be written as

$$h_{\mu\nu}(t, \mathbf{x}) = A_{\mu\nu} \cos(\mathbf{k} \cdot \mathbf{x} - \omega t). \quad (4)$$

This expression illustrates that gravitational disturbances possess both spatial and temporal structure. The observable signal can therefore be studied through quantities such as amplitude, frequency, phase, duration, and time-frequency evolution.

3 LIGO Detection Principle

LIGO measures gravitational waves using laser interferometry. A passing gravitational wave produces a differential variation in the effective lengths of the interferometer arms.

The dimensionless gravitational-wave strain is expressed as

$$h(t) = \frac{\Delta L(t)}{L}, \quad (5)$$

where L represents the nominal interferometer arm length and $\Delta L(t)$ represents the differential displacement.

The magnitude of the effect is extremely small. Consequently, the observed detector output must be treated as a combination of an astrophysical component and a noise component,

$$d(t) = h(t) + n(t), \quad (6)$$

where $d(t)$ is the measured detector data, $h(t)$ is the physical gravitational-wave signal, and $n(t)$ represents instrumental and environmental noise.

4 Gravitational-Wave Signal Structure

For compact binary systems, gravitational-wave signals exhibit a characteristic evolution commonly described through inspiral, merger, and ringdown phases.

A simplified signal representation is

$$h(t) = A(t) \cos[\phi(t)], \quad (7)$$

where $A(t)$ represents the time-dependent amplitude and $\phi(t)$ represents the phase.

The instantaneous frequency may be represented by

$$f(t) = \frac{1}{2\pi} \frac{d\phi(t)}{dt}. \quad (8)$$

The evolution of frequency and amplitude therefore provides information about the physical source and can also be used as features for computational signal analysis.

5 Instrumental Noise and Signal Separation

A major difficulty in gravitational-wave detection is that transient instrumental disturbances can resemble genuine astrophysical signals in their temporal or frequency-domain structure.

Examples of transient detector disturbances include short-duration glitches whose morphology can overlap with portions of gravitational wave signals.

This creates a classification problem in which the computational system must distinguish between

$$\mathcal{D} = \{\text{Cosmic Signal}, \text{Glitch/Noise}\}. \quad (9)$$

The problem is particularly difficult when astrophysical events are rare compared with instrumental noise.

6 Theoretical Framework Proposed in This Study

The central component of the present work is the formulation of theoretical relations intended to investigate possible connections between gravitational-wave observations and the underlying structure of spacetime.

The following relations are introduced as hypotheses. They do not constitute established physical laws and require mathematical, computational, and observational testing.

6.1 First Theoretical Relation

The first proposed relation is defined as

$$\mathcal{R}_1 = [\text{INSERT YOUR FIRST EQUATION}] \quad (10)$$

where the variables are defined according to the physical quantities introduced in the proposed model.

The purpose of this relation is to investigate whether the proposed dependence between the relevant physical quantities can produce an observable consequence in gravitational-wave data.

6.2 Second Theoretical Relation

The second proposed relation is

$$\mathcal{R}_2 = [\text{INSERT YOUR SECOND EQUATION}] \quad (11)$$

This relation is considered within the same conceptual framework and is intended to examine the relationship between the proposed physical parameters and measurable signal characteristics.

6.3 Combined Theoretical Framework

The relations may be considered together through a general functional representation,

$$\mathcal{F} = f(\mathcal{R}_1, \mathcal{R}_2, h(t), f(t), n(t)). \quad (12)$$

The purpose of this formulation is to establish a bridge between the theoretical model and quantities that can potentially be investigated using gravitational-wave observations.

7 Connection with LIGO Data

The theoretical framework can be examined through observable signal properties extracted from LIGO data.

Relevant quantities include:

- Signal-to-noise ratio (SNR);
- Peak frequency;
- Signal duration;
- Amplitude evolution;
- Frequency evolution;
- Time-frequency structure;
- Statistical properties of the measured waveform.

A simplified feature representation can therefore be written as

$$\mathbf{x} = [\text{SNR}, f_{\text{peak}}, T, A, \sigma, \dots]. \quad (13)$$

The relationship between theoretical predictions and these observable quantities provides a possible pathway for empirical testing.

8 Computational Signal Classification

The computational component of this study investigates whether machine-learning methods can distinguish gravitational-wave signals from instrumental noise.

A Random Forest classifier was evaluated using a dataset characterized by a strong class imbalance. The test set contains approximately

$$N_{\text{noise}} = 1582 \quad (14)$$

noise/glitch instances compared with

$$N_{\text{signal}} = 12 \quad (15)$$

cosmic-signal instances.

This imbalance has a direct influence on classification metrics and must therefore be considered when interpreting the apparent overall accuracy of the model.

Table 1: Class-level performance of the evaluated classifier.

Class	Support	Precision	Recall	F1-Score
Glitch / Noise	1582	1.00	0.98	0.99
Cosmic Signal	12	0.21	0.75	0.33

9 Performance Analysis

The results demonstrate a substantial difference between the two classes. The model performs strongly on the majority noise class, while the precision associated with the rare cosmic-signal class is considerably lower.

The result should therefore not be interpreted solely through overall accuracy. The extreme imbalance means that class-specific metrics are more informative for evaluating the ability to identify rare astrophysical events.

10 Root Cause Analysis

Three principal factors may explain the observed classification behavior:

10.1 Morphological Overlap

Certain instrumental glitches can exhibit temporal and spectral structures that resemble portions of astrophysical gravitational-wave signals. This creates overlap between the feature distributions of the two classes.

10.2 Feature-Space Limitations

Basic features such as SNR, peak frequency, and duration provide useful summaries but may not contain sufficient information to represent the full morphology of a gravitational-wave waveform.

10.3 Limited Positive Samples

Only a very small number of positive cosmic-signal samples are available in the evaluated dataset. Consequently, synthetic oversampling methods such as SMOTE may expand the numerical representation of the minority class without introducing new physically independent

gravitational-wave morphologies.

11 Data-Centric Extension

The computational analysis suggests several directions for improving the connection between theoretical signal structure and machine learning.

11.1 Advanced Feature Engineering

Additional temporal and spectral properties may be extracted, including higher-order statistical moments and time-frequency representations. Examples include:

$$X_1 = \text{SNR} \times T, \quad (16)$$

and

$$X_2 = \frac{f_{\text{peak}}}{T}. \quad (17)$$

Wavelet transforms and Constant-Q Transform representations may also provide richer descriptions of waveform morphology.

11.2 Physics-Based Signal Injection

Instead of relying exclusively on synthetic feature-space oversampling, simulated gravitational-wave waveforms can be injected directly into real detector background noise:

$$d_{\text{sim}}(t) = h_{\text{sim}}(t) + n_{\text{real}}(t). \quad (18)$$

This approach preserves realistic detector-noise characteristics while increasing the number of physically motivated positive examples.

11.3 Anomaly Detection

Given the abundance of noise observations, the classification problem may also be formulated as an anomaly-detection problem. A model can first learn the baseline distribution of instrumental noise, $p(n)$, and subsequently identify statistically significant deviations from this baseline as candidate events.

11.4 Decision Threshold

For probabilistic classifiers, the decision rule can be represented as:

$$\hat{y} = \begin{cases} 1, & P(y = 1 \mid \mathbf{x}) \geq \tau, \\ 0, & P(y = 1 \mid \mathbf{x}) < \tau. \end{cases} \quad (19)$$

The threshold τ can be investigated experimentally to study the trade-off between false alarms and missed astrophysical events.

12 Theoretical Interpretation

The theoretical and computational components of this study are connected through the assumption that measurable signal morphology contains information about the physical process responsible for the observed event.

The theoretical framework therefore provides a physical interpretation for quantities used in computational analysis, while the computational analysis provides a possible mechanism for testing whether theoretical differences are reflected in observed data.

This connection can be summarized as:

$$\textit{Physical Model} \longrightarrow \textit{Predicted Signal} \longrightarrow \textit{Observed Data} \longrightarrow \textit{Statistical Test}$$

13 Limitations

The theoretical relations introduced in this study remain hypotheses and should not be interpreted as experimentally established results. Further mathematical derivation is required to establish their consistency with General Relativity and other established physical constraints.

In addition, the computational analysis is limited by the small number of available cosmic-signal examples and the strong imbalance between signal and noise classes.

Consequently, future validation should include larger datasets, simulated waveform injections, numerical experiments, and comparison with real LIGO observations.

14 Conclusion

This study presents a theoretical and computational framework for investigating gravitational-wave signals detected by LIGO.

The framework begins with General Relativity and the description of gravitational waves as perturbations of spacetime, then connects these concepts to detector strain, signal morphology, instrumental noise, and machine-learning classification.

The proposed theoretical relations are formulated as hypotheses that can be examined through mathematical derivation, numerical simulation, and observational data.

The computational analysis further demonstrates that rare-event classification is strongly affected by the underlying data distribution, emphasizing the importance of physically meaningful signal representations and realistic data generation.

Future work will focus on formal derivation of the proposed relations, physics-based waveform injection, advanced time-frequency features, and empirical testing against gravitational-wave observations.

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